

UME311 : Thermodynamics

L	T	P	Cr
3	1	0	3.5

Course Objective: This course introduces basic concepts of thermodynamics and their applications in solving engineering problems. This course also introduces the concept of Rankine cycle, reheat and regeneration.

Syllabus

Basic Concepts: Thermodynamics system & properties, Various processes, Thermodynamic equilibrium, Process: Flow and non flow process, Cycle concept of work and heat, Specific heats, Energy and its form

First Law of Thermodynamics: Concept of internal energy & enthalpy, Energy equation as applied to a close and open system, PMMI, Transient flow processes.

Second Law of Thermodynamics & its Corollaries: Kelvin Plank and Clausius statements, Reversible and Irreversible processes, Carnot cycle, Clausius theorem and concept of entropy, Principle of increase of entropy, PMM2, Thermodynamic temperature scale, Second law analysis of control volume, Availability, Irreversibility, Availability for open and closed system & second law efficiency.

Properties of pure substance: Thermodynamic diagrams, Triple point, Steam tables and their use, Moiller Diagram.

Thermodynamic Relations: Maxwell & T-ds equations.

Vapor Power Cycles: Rankine cycle and modified Rankine cycle, reheat, regenerative and binary cycles.

Course Learning Objectives (CLO)

The students will be able to:

1. analyze and solve problems related to closed systems and steady-flow devices by applying the conservation of energy principle
2. analyze the second law of thermodynamics for various systems and to evaluate the performance of heat engines, refrigerators and heat pumps
3. estimate vapor-liquid properties and solve basic problems using steam tables, Mollier diagrams and equation of state
4. apply the first and second laws of thermodynamics for the complete thermal analysis of vapor power cycle.

Text Books

1. Sonntag, R.E., Borgnakke, C. and Van Wylen, G.J., Fundamentals of Thermodynamics, John Wiley (2007) 6th ed.
2. Nag, P.K., Engineering Thermodynamics, Tata McGraw Hill (2008) 3rd ed.

Reference Books

1. Rao, Y.V.C., Thermodynamics, Universities Press (2004).
2. Ratha Krishana , E., Fundamentals of Engineering Thermodynamics, Prentice Hall of India (2005) 2nd ed.
3. Cengel, Y. A. and Boles, M., Thermodynamics: An Engineering Approach, Tata McGraw Hill (2008).
4. Rogers, G. and Mayhew, Y., Engineering Thermodynamics, Pearson Education (2007) 4th ed.

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	25